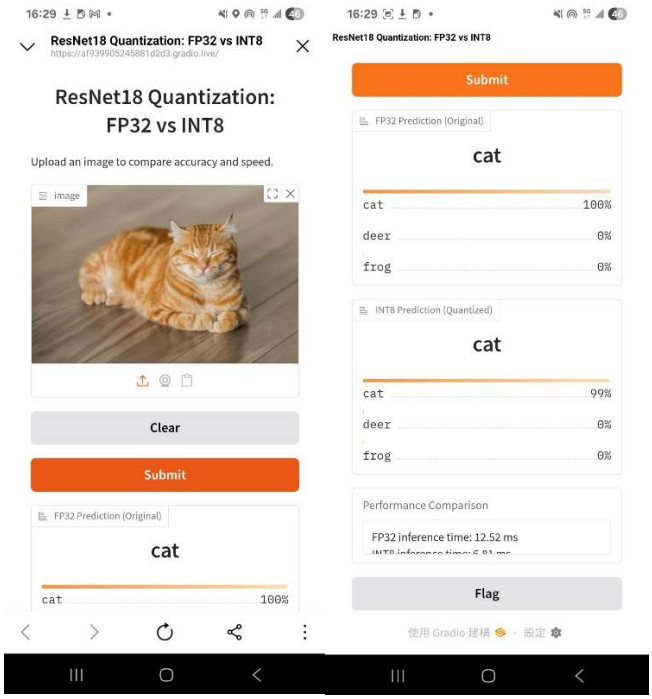
Lab 5 Report

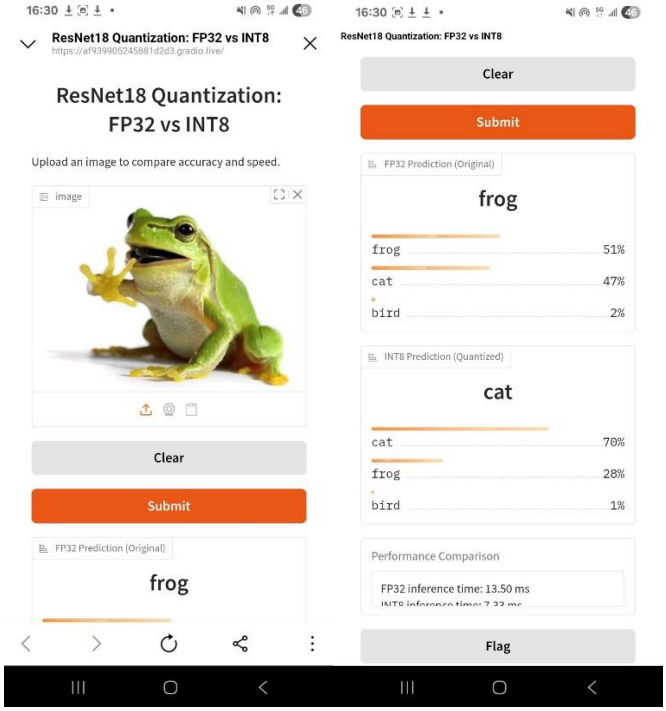
1.

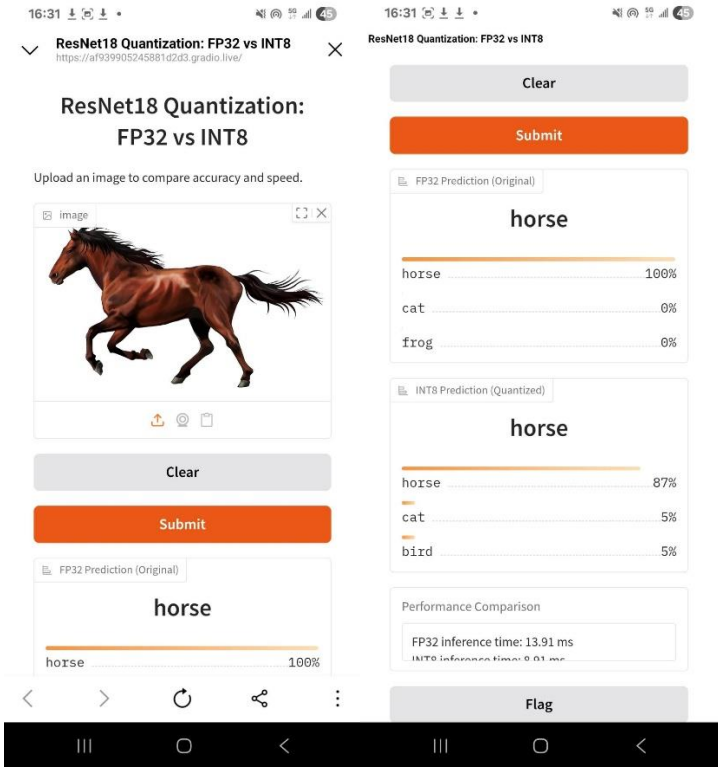
公開網址，以下都是在手機端部署

```
Colab notebook detected. This cell will run indefinitely so that you can see errors and logs. To turn off, set debug=False in launch().
* Running on public URL: https://af939905245881d2d3.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the working directory to deploy to Hugging Face Space
```







Gradio UI 設計:

```
# ====== ONNX Sessions ======
providers = ort.get_available_providers()

sess_int8 = build_session(MODEL_PATH_INT8, providers=providers)
in_int8 = sess_int8.get_inputs()[0].name
out_int8 = sess_int8.get_outputs()[0].name
```

此部分透過 ONNX Runtime 建立兩個獨立的推論 Session，分別對應量化前的 FP32 模型與量化後的 INT8 模型。這讓我們能在相同輸入下直接比較兩者的推論結果與效能差異。

讓使用者比較深度學習模型在不同精度下的 inference time 和 Accuracy

```
# Your progarm
# FP32 Inference
start_t = time.time()
res_fp32 = sess_fp32.run([out_fp32], {in_fp32: x})[0]
fp32_ms = (time.time() - start_t) * 1000

# INT8 Inference
start_t = time.time()
res_int8 = sess_int8.run([out_int8], {in_int8: x})[0]
int8_ms = (time.time() - start_t) * 1000
```

本段程式同時對 FP32 與 INT8 模型執行推論，並以毫秒計算單張影像的 inference latency。透過 softmax 後的 top-3 結果，可量化量化前後的準確度差異，並由兩者推論時間估算 speedup。

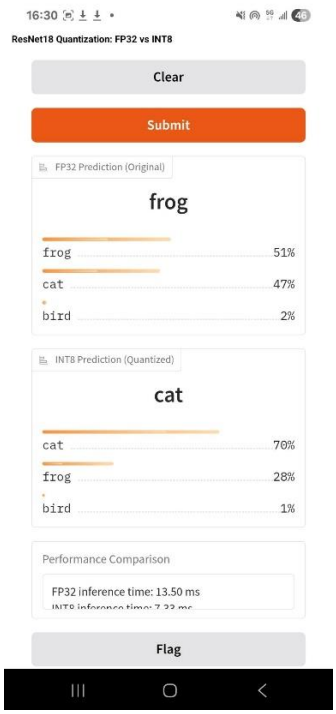
```
# ====== Gradio UI ======
# TODO : Building GUI Interface
demo = gr.Interface(
        fn = compare_fp32_int8,
        inputs = gr.Image(type="pil"),
        outputs = [
                gr.Label(label="FP32 Prediction (Original)", num_top_classes=3),
                gr.Label(label="INT8 Prediction (Quantized)", num_top_classes=3),
                gr.Textbox(label="Performance Comparison")
        ],
        title = "ResNet18 Quantization: FP32 vs INT8",
        description = "Upload an image to compare accuracy and speed."

)
```

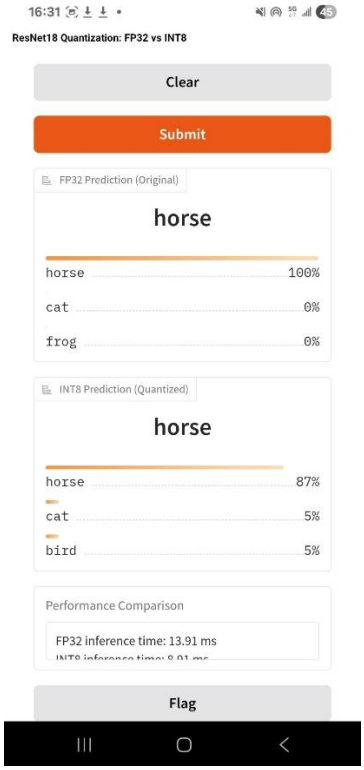
本系統採用 Gradio 建立可視化前端，用戶能直接上傳影像，查看 FP32 與 INT8 模型的推論結果與推論時間。這使量化效益可視化，便於展示量化後的效能改善。

2.

在 inference time 上，INT8 的 inference time 比 FP32 的快，但在 accuracy 上 FP32 精準蠻多，像 frog INT8 model 竟然說是 cat (第一題第二個)



在 horse 的判斷上，FP32 精準的說出是 horse，但 INT8 在其他的 label 還是會有些趴數



3.

出得很好，非常棒，很好玩的一個實驗，能練習到真的部署在裝置